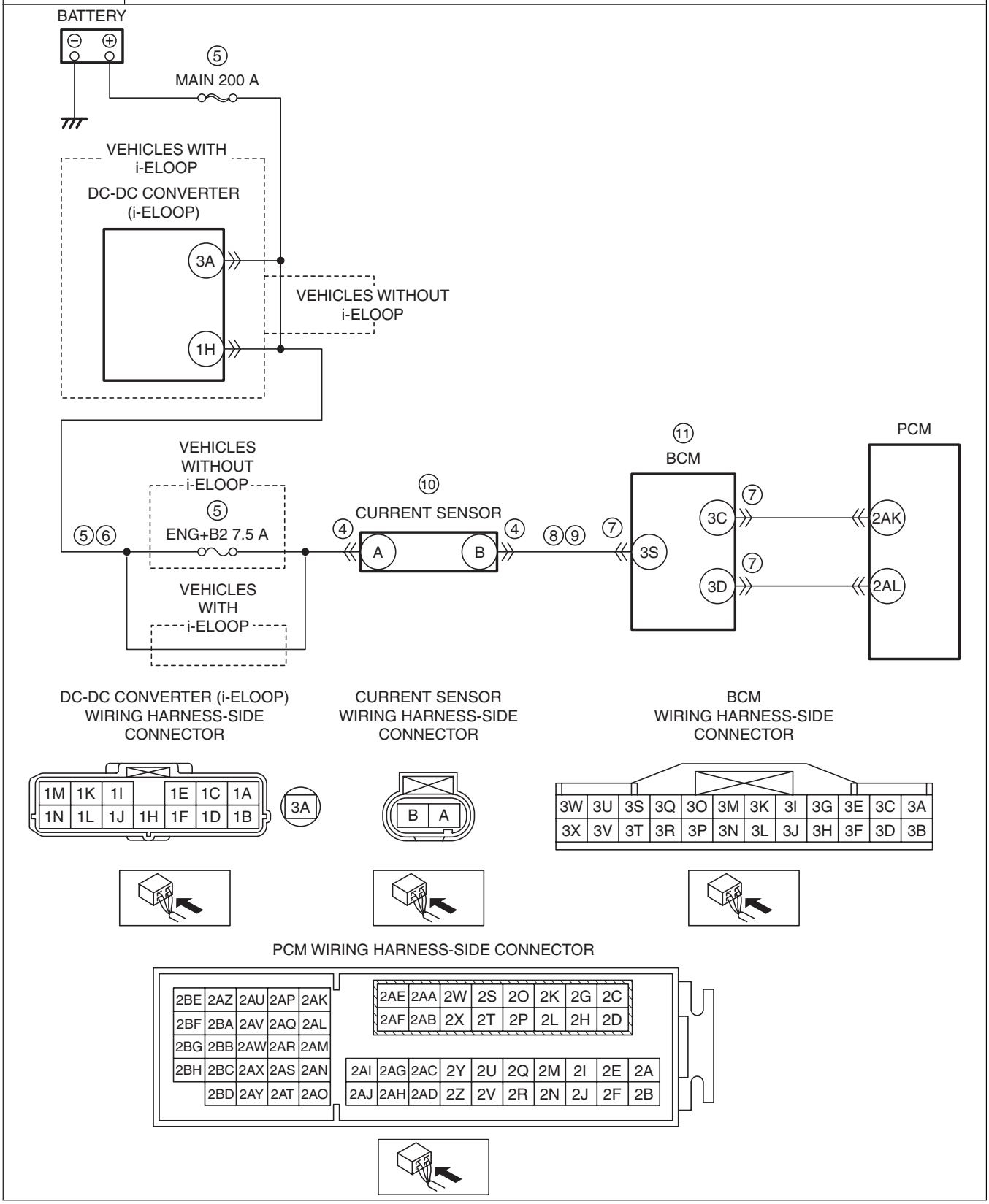
DTC U1007:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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DTC U1007:00	CAN/LIN communication system: current sensor information communication error with BCM
DETECTION CONDITION	• PCM detects a current sensor information communication error from BCM. **Diagnostic support note** • This is a continuous monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA is not available. • Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Inhibits engine-stop by operating the i-stop function. • Inhibits a part of the generator output control.
POSSIBLE CAUSE	• Communication line malfunction between current sensor and BCM • Communication line malfunction between BCM and PCM • Current sensor connector or terminals malfunction • Short to ground or open circuit in current sensor power supply circuit (without i-ELOOP) — Short to ground in wiring harness between MAIN 200 A fuse and current sensor terminal A — MAIN 200 A fuse and/or ENG+B2 7.5 A fuse malfunction — Open circuit in wiring harness between battery positive terminal and current sensor terminal A • Short to ground or open circuit in current sensor power supply circuit (with i-ELOOP) — Short to ground in wiring harness between DC-DC converter (i-ELOOP) terminal 1H and current sensor terminal A — Open circuit in wiring harness between DC-DC converter (i-ELOOP) terminal 1H and current sensor terminal A • BCM connector or terminals malfunction • Short to ground in wiring harness between current sensor terminal B and BCM terminal 3S • Open circuit in wiring harness between current sensor terminal B and BCM terminal 3S • Current sensor malfunction • BCM malfunction

DTC U1007:00	CAN/LIN communication system: current sensor information communication error with BCM



Repeatability Verification Procedure

1. Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
2. Start the engine.

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
4	**INSPECT CURRENT SENSOR CONNECTOR CONDITION** **Note** • Always disconnect current sensor connector before disconnecting the negative battery cable. • Switch the ignition off. • Remove the service plug. (with i-ELOOP) (See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].) • Disconnect the current sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 12.
		No	Vehicles with i-ELOOP: • Go to Step 6. Vehicles without i-ELOOP: • Go to the next step.

STEP	INSPECTION		ACTION
5	**INSPECT CURRENT SENSOR POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the current sensor connector is disconnected. • Measure the voltage at the current sensor terminal A (wiring harness-side). • Is the voltage **B+**?	Yes	Go to Step 7.
		No	Inspect the MAIN 200 A fuse and ENG+B2 7.5 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200 A fuse and current sensor terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the malfunctioning fuse. • If the fuse is damaged: — Replace the malfunctioning fuse. • If all fuses are normal: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and current sensor terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 12.
6	**INSPECT CURRENT SENSOR POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the current sensor connector is disconnected. • Measure the voltage at the current sensor terminal A (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 1H and current sensor terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short or open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short or open circuit. Go to Step 12.
7	**INSPECT BCM CONNECTOR CONDITION** • Disconnect the BCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 12.
		No	Go to the next step.

STEP	INSPECTION		ACTION
8	**INSPECT CURRENT SENSOR LIN COMMUNICATION LINE FOR SHORT TO GROUND** • Verify that the current sensor and BCM connectors are disconnected. • Inspect for continuity between current sensor terminal B (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between current sensor terminal B and BCM terminal 3S. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 12.
		No	Go to the next step.
9	**INSPECT CURRENT SENSOR LIN COMMUNICATION LINE FOR OPEN CIRCUIT** • Verify that the current sensor and BCM connectors are disconnected. • Inspect for continuity between current sensor terminal B (wiring harness-side) and BCM terminal 3S (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between current sensor terminal B and BCM terminal 3S. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 12.
10	**INSPECT CURRENT SENSOR** • Inspect the current sensor. (See CURRENT SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the current sensor, then go to Step 12. (See CURRENT SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
11	**INSPECT BCM** • Inspect the BCM. (See BODY CONTROL MODULE (BCM) INSPECTION.) • Is there any malfunction?	Yes	Replace the BCM, then go to the next step. (See BODY CONTROL MODULE (BCM) REMOVAL/ INSTALLATION.)
		No	Go to the next step.
12	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Reinstall the service plug. (with i-ELOOP) (See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].) • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1.
		No	Go to the next step.
13	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.